

12.1 Unit description

Introduction

This unit requires that students plan an experiment, carry out an experiment, record measurements, analyse their own results and draw conclusions.

This unit may be completed at any time during the second year of the course but it would be more appropriate to administer this assessment near the end of the A2 year. This assessment should take no more than 2 hours to complete.

All candidates within one class may produce a plan for the same experiment as each other and do the same practical work; however it is vital that candidates demonstrate that the assessed work that they produce is entirely their own work.

If more than one class of students take this assessment at different times, then the groups must submit different plans for assessment to prevent plagiarism.

Candidates' work may be based on briefing material provided by Edexcel or briefing material that is devised by the centre. The brief for this assessment is to be set by the teacher, not the student; however, briefs should reflect the interests of students where possible.

Planning component

The planning component of this assessment may be done at a different time to the other components. Plans produced by the students may be based on either a briefing provided by the centre or a briefing provided by Edexcel.

Experiment and analysis of results

The experiment and analysis of results may be based on the plan produced by each individual student in the first part of this assessment or it may be based on a plan that is provided by Edexcel or a plan that is devised by the centre. If the centre produces the plan on which the experiment is based, it is vital that the plan provides the opportunity for students to achieve the full range of marks that are available. Centre devised plans should contain some flaws so that students are able to modify the experiment while they are doing it and suggest improvements. Centre-devised plans should ensure that a non-linear relationship exists between the variables that are investigated.